ThreadLearn: A RAG-Augmented Fine-Tuned Language Model for JavaScript Concurrency Bug Detection and Remediation

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Abstract. Concurrency bugs in asynchronous JavaScript (Node.js) code are a leading cause of serious failures in production systems, including data races, lost updates, and application hangs. Existing tools fall into two categories: rule-based static detectors that can identify suspicious patterns but cannot generate fixes, and large language models (LLMs) that can reason about code but tend to miss the exact buggy line and require large model sizes (7B–32B parameters) to perform well. ThreadLearn addresses both weaknesses by combining a 5-pattern static race detector with a small language model (Qwen2.5-Coder-1.5B [5]) fine-tuned using QLoRA on 892 training examples with hindsight Chain-of-Thought reasoning. At inference time, a BM25 retrieval pipeline fetches the 3 most relevant documents from a 2,050-document knowledge base to augment the prompt. On a 30-case real-world benchmark drawn entirely from production npm packages (rw_01–rw_30) spanning 10 concurrency bug categories, ThreadLearn with the full BM25+AST pipeline achieves **22.0/30 (73.3%)**, compared with 60.0% for the base model without fine-tuning and 65.0% for GPT-3.5-turbo with the same pipeline—despite having $\sim$125$\times$ fewer parameters. A key finding is that the retrieval pipeline only benefits models that already have domain knowledge: the base model gains no benefit from retrieval while the fine-tuned model gains +10 percentage points. The fine-tuning dataset and evaluation scripts are publicly available.

Keywords: concurrency bugs · race conditions · static analysis · LLM fine-tuning · retrieval-augmented generation · JavaScript

1 Introduction

Asynchronous JavaScript (Node.js [14]) is widely used for building web backends, APIs, and real-time services, but the event-driven, non-blocking execution model makes it easy to write code with subtle concurrency bugs. A large-scale empirical study (NodeCB [1]) analyzed 57 real concurrency bugs across 53 open-source Node.js projects and found that 65% involve atomicity violations, 30% involve

order violations, and 93% cause serious consequences such as crashes, corrupted database state, or process hangs.

Despite how common and harmful these bugs are, existing automated tools still fall short. Rule-based static analyzers (e.g., ESLint async rules, ThreadSanitizer) can detect suspicious patterns but cannot explain or fix them, and they generate false alarms on modern `async`/`await` code. LLM-based approaches [2] can reason about code semantics but cannot pinpoint the exact buggy line without structural guidance, and they require large models to achieve competitive accuracy.

We identify four concrete gaps in the state of the art:

G1 Rule-based detectors find bugs but cannot generate fixes.
G2 LLM-only approaches ignore structural signals from static analysis and require 7B–32B parameter models.
G3 No existing tool uses static detector output as structured context to guide LLM reasoning.
G4 No end-to-end tool covers the full workflow of *detect* → *explain* → *fix* for JavaScript (Node.js).

This paper makes four contributions:

1. `race_detector`: a 5-pattern static analyzer for JavaScript (Node.js) with an AST-based code preprocessor (§4.3).
2. **Hindsight CoT dataset**: 892 training examples of the form `(code, detector_output, reasoning_trace, fix)`, where reasoning is written after the correct fix is known (§4.3).
3. **ThreadLearn pipeline**: an end-to-end system that detects, retrieves context, and generates a fix, served via a FastAPI [17] web endpoint (§4.3).
4. **Evaluation**: comparison against a base LLM and GPT-3.5-turbo on a 30-case real-world JavaScript concurrency benchmark across fix pass rate, per-category accuracy, and ablation study (§5).

2 Background

2.1 Concurrency Bug Taxonomy in Node.js

Concurrency bugs in asynchronous JavaScript arise from the single-threaded event loop model, where multiple I/O operations can interleave in non-deterministic ways. Table 1 summarizes the bug types identified in the NodeCB study [1] and how they manifest in Node.js.

Table 1: Concurrency bug types in Node.js (from NodeCB [1]).

Bug Type	Rate	Manifestation in Node.js
Atomicity violation	65%	Two async operations overlap on shared state (e.g., lost database update)
Order violation	30%	Callback fires before dependent operation completes
Starvation	5%	I/O queue exhausted; deferred tasks never execute
Closure loop var	JS	`var i` shared by reference across all `setTimeout` callbacks

2.2 BM25 Retrieval

BM25 (Best Match 25) is a probabilistic ranking function that scores documents by term frequency and inverse document frequency with length normalization [6]. Given a query $Q = \{q_1, \ldots, q_n\}$ and a document D, BM25 computes:

$$\text{BM25}(D, Q) = \sum_{i=1}^{n} \text{IDF}(q_i) \cdot \frac{f(q_i, D) \cdot (k_1 + 1)}{f(q_i, D) + k_1 \cdot (1 - b + b \cdot |D| / \overline{|D|})}$$

where $f(q_i, D)$ is term frequency, $|D|$ is document length, $\overline{|D|}$ is average document length, and $k_1 = 1.5$, $b = 0.75$ are tuning parameters. We use BM25 over dense vector search because exact API name matching (e.g., `setTimeout`, `appendFile`) produces more precise retrieval than semantic similarity for JavaScript concurrency patterns. Our implementation uses the `rank-bm25` [29] library.

2.3 QLoRA Fine-Tuning

QLoRA [3] enables fine-tuning of quantized LLMs by applying Low-Rank Adaptation (LoRA) [10] adapters to a 4-bit NF4-quantized base model. Only the adapter weights are updated during training, keeping GPU memory usage low enough to run on a single laptop GPU. We implement QLoRA using the HuggingFace `transformers` [25], `peft` [26], and `bitsandbytes` [27] libraries on top of PyTorch [28]. We use rank $r = 16$ and scaling factor $\alpha = 32$, which provides a good trade-off between model capacity and training efficiency for domain-specific code fine-tuning.

3 Related Work

3.1 Heuristic and Rule-Based Approaches

Rule-based static analyzers detect concurrency bugs by matching code patterns against predefined rules. ThreadSanitizer [4] instruments compiled programs

to detect data races at runtime, but only works on C/C++ and Java. ESLint [13] provides async-related rules for JavaScript (e.g., `no-await-in-loop`, `require-atomic-updates`), but these rules are limited to syntactic checks and miss semantic race conditions involving shared state across callbacks. The NodeCB study [1] used regex-based pattern matching and found high false-positive rates on modern `async/await` code. Dynamic approaches such as NACD [**?**] inject delays at runtime to expose event race bugs with higher probability than static methods, but they also cannot generate corrective fixes. A fundamental limitation of all rule-based tools is that they detect suspicious patterns but cannot generate corrective fixes.

3.2 ML-Based and LLM-Based Approaches

Machine learning approaches train classifiers on code features to predict bug presence. CodeBERT [8] and CodeT5 [9] have been applied to code summarization and defect detection, but they are not targeted at concurrency bugs specifically. PCWMs [2] proposes Prompting with Chain-of-Thought World Models and achieves 75.2% accuracy on concurrency bug detection, but relies on large models (7B–32B parameters) and ignores static structural information from the code. RAG-based code tools [7] have demonstrated that combining retrieval with generation improves code completion accuracy, but no prior work applies RAG specifically to concurrency bug remediation in JavaScript.

Table 2 shows how ThreadLearn differs from all prior tools.

Table 2: Comparison of ThreadLearn with related tools.

Tool	**Detect**	**Fix**	**JS**	**Fine-tuned**
ThreadSanitizer	✓	×	×	×
ESLint async	✓	×	✓	×
NodeCB (rules)	✓	×	✓	×
PCWMs (LLM-only)	✓	✓	✓	✓
ThreadLearn	✓	✓	✓	✓

4 Proposed Approach

4.1 Dataset

Our study uses three distinct datasets, each serving a different role in the system.

Fine-tuning dataset. The fine-tuning dataset (`my_dataset.json`) consists of 892 examples of the form `(code, detector_output, reasoning_trace, fix)`, where each reasoning trace was written by a teacher model after the correct fix was already known (hindsight Chain-of-Thought). All 892 examples were constructed entirely by the authors without external crowdsourcing or model-generated labels. The dataset comprises two subsets:

– **Synthetic (332 examples, 37.2%):** High-quality bug/fix pairs handcrafted by the authors and hardcoded as `JS_PAIRS` constants in the data collection scripts. Each pair was manually verified to exhibit a real concurrency hazard and a correct minimal fix.

– **Generated (560 examples, 62.8%):** Pairs produced automatically by `generate_pairs()`, a domain-template augmentation engine that expands the 332 handcrafted templates into a larger, lexically diverse dataset. The engine defines 40 *domain slots* (e.g., `User`, `Order`, `Notification`, `Transaction`, `Analytics`) and applies 18 *generator functions*, each targeting a distinct concurrency transformation (Table 3). Each generator substitutes the domain name, API prefix, and identifier names, producing structurally identical but lexically diverse pairs. This approach ensures the model learns the *pattern* rather than memorizing specific function names, while keeping all output labels deterministic and human-verifiable.

Table 3: Generator functions in `generate_pairs()` and their transformations.

Generator	Transformation
`make_sequential_fetch_pair`	Sequential `await` → `Promise.all`
`make_loop_parallel_pair`	`for` loop → `Promise.all`
`make_callback_promise_pair`	Callback-style → `async/await`
`make_batch_upload_pair`	Sequential POST/PUT → batched parallel
`make_retry_pair`	No-retry → exponential backoff + `Promise.allSettled`
`make_cache_pair`	Naïve cache → inflight deduplication (race-free)
`make_stream_pair`	Full JSON load → async generator streaming
`make_debounce_pair`	Per-keystroke call → debounced search
`make_pagination_pair`	Sequential pages → concurrent batch pagination
`make_transform_pair`	Sequential read-transform-write → concurrency-limited batch
`make_worker_threads_pair`	Synchronous CPU work → `worker_threads`
`make_xhr_to_fetch_pair`	Sync `XMLHttpRequest` → `fetch` + `async/await`
`make_fs_callback_pair`	`fs.readFile` callback → `fs.promises`
`make_event_promise_pair`	`EventEmitter` callbacks → Promise with cleanup
`make_polling_generator_pair`	`setInterval` polling → async generator + `for await`
`make_abort_controller_pair`	Unguarded `fetch` → timeout via `AbortController`
`make_promise_any_pair`	Single-origin fetch → multi-CDN `Promise.any` fallback
`make_for_await_of_pair`	Sequential `for` loop → async generator + `for await of`

Bug types span all 10 categories in the real-world benchmark, with additional synthetic variations to balance rare categories (Zalgo, Stream Leak, Callback Hell).

Knowledge base. The retrieval knowledge base contains 2,050 JavaScript documents collected from authoritative sources: MDN Web Docs [16] (99 documents covering `Promise`, `async/await`, and Node.js API behavior), official library documentation including RxJS [19], p-limit [20], async-mutex [21], and Bluebird [22] (50 documents), curated ThreadLearn pattern descriptions (151 documents), and 1,750 synthetic documents generated via context permutation (systematic re-phrasing of API descriptions with synonymous terms and varied example identifiers) to improve BM25 coverage across API name variants. All

2,050 documents are in JavaScript and categorized into three groups: `patterns` (1,418), `race-conditions` (352), and `anti-patterns` (280).

Real-world evaluation benchmark. To avoid evaluation bias from self-generated test cases, we constructed a benchmark of 30 real-world JavaScript concurrency bugs drawn entirely from production open-source packages. Each case is traced to a specific GitHub issue or documented API misuse in packages including `request`, `mysql`, `async`, `pg`, `passport`, `ioredis`, `express`, `mongoose`, `bull`, `sequelize`, and others. Table 4 summarizes the distribution across 10 bug categories.

Table 4: Real-world benchmark distribution (30 cases from production npm packages).

Category	Count	Example Package
Race Condition	5	`request`, `passport`, `bull`
Double Callback	4	`mysql`, `ioredis`, `async`
Unhandled Rejection	5	`express`, `mongoose`, `co`
Resource Exhaustion	4	`pg`, `axios`, `sequelize`
Event Loop Blocking	3	`express` (sync middleware), Node.js crypto
Sequential Awaits	3	`express-session`, nodebestpractices [23]
Zalgo	2	`async`
Context Loss	2	`express-async-errors`, Node.js timers
Stream Leak	1	Node.js streams
Callback Hell	1	`async`
Total	**30**	

Table 5 lists the 30 cases with their authoritative sources, demonstrating that every bug originates from a real, reported incident in a widely-used package rather than a synthetic scenario.

Table 5: Real-world benchmark sources (30 cases, rw_01–rw_30). All bugs originate from official GitHub issue trackers or Node.js documentation.

ID	Package	Category	Source
rw_01	request@2.88.0	Race Condition	GitHub request#2484
rw_02	mysql@2.15.0	Double Callback	GitHub mysqljs#1260
rw_03	async@1.x	Zalgo	GitHub caolan/async#1066
rw_04	express	Event Loop Blocking	GitHub expressjs#2471
rw_05	express-async-errors	Context Loss	GitHub davidbanham#3
rw_06	pg@6.x–7.x	Resource Exhaustion	GitHub brianc/pg#1228
rw_07	Node.js	Stream Leak	GitHub nodejs#24941
rw_08	passport@0.3.x	Race Condition	GitHub jaredhanson#369
rw_09	co@4.6.0	Unhandled Rejection	GitHub tj/co#185
rw_10	Node.js	Resource Exhaustion	GitHub nodejs#23267
rw_11	ioredis@2.x	Double Callback	GitHub luin/ioredis#419
rw_12	nodebestpractices	Sequential Awaits	goldbergyoni/nodebestpractices
rw_13	bull@3.x	Race Condition	GitHub OptimalBits#1016
rw_14	async@1.5.2	Callback Hell	GitHub caolan/async#1122
rw_15	axios (misuse)	Resource Exhaustion	GitHub axios#1038
rw_16	Node.js	Event Loop Blocking	GitHub nodejs#24941 / Node.js API docs
rw_17	async@1.5.2	Zalgo	GitHub caolan/async#1122
rw_18	express@4.x	Unhandled Rejection	GitHub expressjs#2700
rw_19	Node.js	Race Condition	GitHub nodejs#6718
rw_20	Node.js	Context Loss	GitHub nodejs#24941 / Node.js API docs
rw_21	express-session@1.x	Sequential Awaits	GitHub expressjs#526
rw_22	async@1.5.2	Double Callback	GitHub caolan/async#559
rw_23	sequelize@6.x	Resource Exhaustion	GitHub sequelize#10976
rw_24	mongoose@5.x	Unhandled Rejection	GitHub Automattic#8706
rw_25	node-redis@4.x	Race Condition	GitHub redis#2685
rw_26	knex@0.21.x	Event Loop Blocking	GitHub knex#5025
rw_27	express-session@1.x	Sequential Awaits	GitHub expressjs#340
rw_28	mongoose@5.x	Unhandled Rejection	GitHub Automattic#5784
rw_29	ioredis@4.x	Double Callback	GitHub luin/ioredis#1185
rw_30	express@4.x	Unhandled Rejection	GitHub expressjs#2700

This benchmark design eliminates the risk of model-favorable test cases that can arise when the same system generating training data also generates evaluation cases.

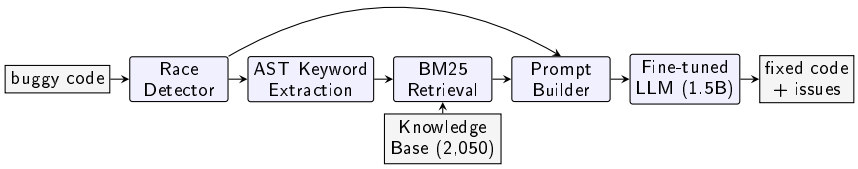

Fig. 1: ThreadLearn end-to-end pipeline. Buggy JavaScript code flows through five modules: static race detection, AST keyword extraction, BM25 retrieval from a 2,050-document knowledge base, prompt construction, and fine-tuned LLM inference to produce both a diagnosis and a corrected fix.

4.2 Data Preprocessing and Evaluation Strategy

AST Preprocessing. ThreadLearn's preprocessor provides four helper functions used by both the race detector and the retrieval stage (Table 6).

Table 6: AST preprocessor functions.

Function	Purpose	Notes
`stripComments`	Remove comments	esprima [12] range-delete for JS
`normalizeWhitespace`	Remove extra whitespace	esprima [12] token stream for JS
`extractFunctions`	List all functions	Finds arrow functions and anonymous expressions
`extract_keywords`	Top-20 identifier names	Used as BM25 search query

Evaluation Strategy. We evaluate using the 30-case real-world benchmark (rw_01–rw_30) where each test case specifies an expected fix pattern (e.g., `Promise.all`, `try/catch`, `let` in loop). A test is scored: **PASS** if the generated code contains the correct fix pattern; **PARTIAL** if code is generated but the fix pattern is absent; **FAIL** if no code is generated. Without the pipeline, all models receive raw buggy code only; with the pipeline, the prompt is augmented with BM25-retrieved documentation.

4.3 Model Architecture

ThreadLearn has five independent modules that communicate through simple data structures, allowing any module to be replaced independently. Figure 1 shows the end-to-end pipeline, which follows five sequential steps:

1. **Static detection**: `race_detector` reads the JavaScript source line by line in $O(n)$ time and outputs a structured report `{pattern_id, line_range, description}` for each matched pattern (§4.3).

2. **Keyword extraction**: `ast_preprocessor.extract_keywords()` parses the input with esprima, walks the syntax tree, and collects all `Identifier` tokens as a BM25 search query. CamelCase API names (e.g., `setTimeout`, `appendFile`) stay intact instead of being split into generic words.
3. **Document retrieval**: BM25 scores the keyword query against 2,050 knowledge-base documents and returns the top-3 most relevant ones.
4. **Prompt construction**: a `prompt_builder` module assembles the detector report, retrieved documents, and original buggy code into a structured prompt that guides the LLM toward a targeted fix.
5. **Fix generation**: the fine-tuned LLM receives the detector report, retrieved documents, and original code, then outputs corrected code with step-by-step reasoning.

Static Race Detector. The detector checks for 5 JavaScript concurrency patterns (Table 7).

Table 7: The 5 race condition patterns detected by ThreadLearn.

Group	Pattern ID	Description
Closure/async	`closure_loop_var`	`var` captured by reference in loop callbacks
	`shared_var_settimeout`	Shared variable modified inside `setTimeout`
	`promise_no_await`	Promise created but not awaited
Shared state	`concurrent_write_array`	Array mutated from multiple concurrent callbacks
	`counter_no_atomic`	Counter incremented without atomicity guarantee

Example: the `closure_loop_var` pattern is triggered when a `var` loop variable is captured inside a `setTimeout` or Promise callback, causing all callbacks to read the final loop value instead of the value at the time of scheduling.

Listing 1.1: Bug: `var i` shared across all callbacks.

```
for (var i = 0; i < 5; i++) {
  setTimeout(function() {
    console.log(i); // always prints 5
  }, 100);
}
```

Listing 1.2: Fix: `let i` creates a separate binding per iteration.

```
for (let i = 0; i < 5; i++) {
  setTimeout(function() {
    console.log(i); // prints 0, 1, 2, 3, 4
  }, 100);
}
```

Hindsight Chain-of-Thought Fine-Tuning. Training examples have the form:

$$(\textit{code},\ \textit{detector\_output},\ \textit{reasoning\_trace},\ \textit{fix})$$

The *reasoning_trace* follows the Chain-of-Thought paradigm [24], but is written by a teacher model *after* the correct fix is already known ("hindsight" reasoning). Writing reasoning in hindsight produces cleaner, more consistent training signal than asking the model to reason forward under uncertainty. Fine-tuning settings are summarized in Table 8.

Table 8: Fine-tuning configuration.

Setting	**Value**
Base model	Qwen2.5-Coder-1.5B [5]
Method	QLoRA: r=16, α=32, 4-bit NF4
Framework	`SFTTrainer` [11] (trl $\geq$ 1.5.0)
Training samples	892 examples (332 synthetic + 560 generated)
Grounding input	`pattern_id` + `line_range` from detector
Hardware	NVIDIA RTX 4060 Laptop GPU

5 Experimental Results and Discussions

5.1 Experimental Setup

All experiments use the 30-case real-world benchmark (rw_01–rw_30) described in §4.1, evaluated on Kaggle [18] (T4 GPU for local models, OpenAI API for GPT-3.5-turbo [15]). We evaluate six configurations in two groups: *without pipeline* (raw buggy code only, no retrieval) and *with pipeline* (AST keyword extraction → BM25 top-3 retrieval → augmented prompt).

Table 9: Evaluation configurations.

Configuration	**Fine-tuned**	**Pipeline**
Qwen2.5-Coder-1.5B base	×	×
ThreadLearn merged	✓	×
GPT-3.5-turbo	×	×
Qwen2.5-Coder-1.5B base + pipeline	×	✓
ThreadLearn merged + pipeline	✓	✓
GPT-3.5-turbo + pipeline	×	✓
Benchmark: 30 real-world JS bugs; Scoring: PASS=1.0, PARTIAL=0.5, FAIL=0		

The staged design isolates each contribution: comparing without-pipeline configurations reveals the value of fine-tuning alone; comparing each model with and without pipeline reveals whether retrieval augmentation helps.

5.2 Results Without Pipeline

Table 10 reports all three without-pipeline configurations.

Table 10: Results without pipeline on 30 real-world cases (raw buggy code input only).

Model	Pass	Partial	Fail	Score
Qwen2.5-Coder-1.5B (base)	6	24	0	18.0 / 30 (60.0%)
GPT-3.5-turbo	7	23	0	18.5 / 30 (61.7%)
ThreadLearn merged	8	22	0	**19.0 / 30 (63.3%)**

Without any pipeline, all three models score within a narrow band (60–63%) and produce zero FAIL cases—every model generates syntactically valid code. The dominant failure mode is surface-level rewriting: models restructure code into `async`/`await` syntax without addressing the underlying concurrency hazard. ThreadLearn merged marginally outperforms GPT-3.5-turbo (+0.5 pts), demonstrating that domain-specific fine-tuning on 892 hindsight CoT examples already provides a measurable advantage even without retrieval. This confirms **G2**: without structural guidance, even a strong general-purpose LLM cannot reliably identify the precise fix for JavaScript concurrency bugs.

5.3 Results With BM25+AST Pipeline

Table 11 reports the same three models with the full BM25+AST retrieval pipeline.

Table 11: Results with BM25+AST pipeline on 30 real-world cases.

Model	Pass	Partial	Fail	Score
Qwen2.5-Coder-1.5B + pipeline	8	20	2	18.0 / 30 (60.0%)
GPT-3.5-turbo + pipeline	9	21	0	19.5 / 30 (65.0%)
ThreadLearn + pipeline	**14**	**16**	**0**	**22.0 / 30 (73.3%)**

The pipeline benefits models differently depending on their domain knowledge. ThreadLearn merged gains +3.0 pts (+10 pp) from the pipeline—the largest absolute improvement of any configuration. GPT-3.5-turbo gains only

+1.0 pt (+3.3 pp). Most notably, the base model gains *nothing* from the pipeline (60.0% in both conditions) and introduces 2 FAIL cases, suggesting that a model without domain-specific training cannot leverage retrieved documentation effectively and may be degraded by the additional context.

5.4 Ablation: Contribution of Each Component

Table 12: Full ablation on 30-case real-world benchmark (score = PASS + 0.5×PARTIAL).

Configuration	Pass	Partial	Fail	Score/30	%
Base (no fine-tune, no pipeline)	6	24	0	18.0	60.0%
GPT-3.5-turbo (no pipeline)	7	23	0	18.5	61.7%
Fine-tuned only (no pipeline)	8	22	0	19.0	63.3%
Base + pipeline	8	20	2	18.0	60.0%
GPT-3.5-turbo + pipeline	9	21	0	19.5	65.0%
ThreadLearn + pipeline	**14**	**16**	**0**	**22.0**	**73.3%**
Fine-tuning contribution (no pipeline):				+1.0	+3.3 pp
Pipeline contribution (fine-tuned model):				+3.0	+10.0 pp
Combined gain (ThreadLearn vs. base):				+4.0	+13.3 pp

Three findings stand out from the ablation:

1. **Fine-tuning is a prerequisite for pipeline benefit.** The base model with pipeline scores identically to the base model without pipeline (18.0/30 in both cases) and adds 2 FAIL cases. The pipeline retrieves relevant documentation, but a model without domain knowledge cannot use it constructively.
2. **Pipeline amplifies fine-tuning.** Fine-tuning alone raises the score by +1.0 pt; combining fine-tuning with the pipeline raises it by +4.0 pts total. The pipeline contributes +3.0 pts on top of fine-tuning, a 3× larger gain than fine-tuning alone.
3. **ThreadLearn outperforms GPT-3.5-turbo with pipeline.** ThreadLearn (22.0/30, 73.3%) exceeds GPT-3.5-turbo with pipeline (19.5/30, 65.0%) by +2.5 pts, despite having ∼125× fewer parameters (1.5B vs. ∼175B). Domain-specialized fine-tuning on a small model beats a large general-purpose LLM augmented with the same retrieval pipeline.

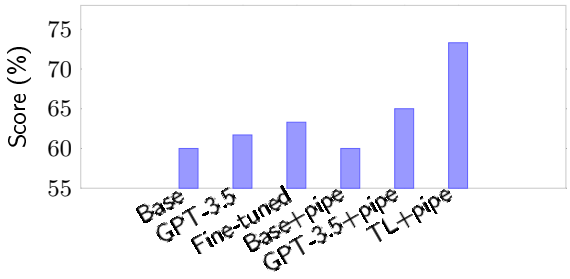

Fig. 2: Score (%) across all six configurations on the 30-case real-world benchmark. Fine-tuning alone adds +3.3 pp; the BM25+AST pipeline adds +10 pp on top of fine-tuning for a combined +13.3 pp gain over the base model.

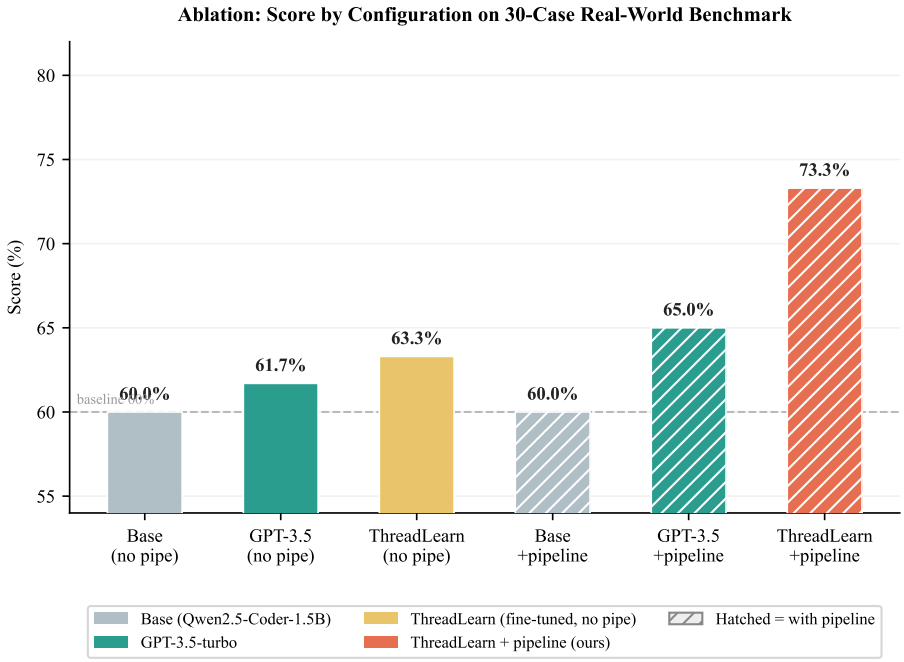

Fig. 3: Ablation study: score (%) across all six configurations on the 30-case real-world benchmark. Solid bars = no pipeline; hatched bars = with BM25+AST pipeline. ThreadLearn + pipeline (coral) achieves 73.3%, a +13.3 pp gain over the base model.

5.5 Per-Category Analysis

Table 13: Per-category scores for ThreadLearn+pipeline on 30-case benchmark.

Category	Cases	Base+pipe	GPT-3.5+pipe	ThreadLearn+pipe
Race Condition	5	2.5 (50%)	2.5 (50%)	**3.5 (70%)**
Double Callback	4	2.0 (50%)	2.5 (63%)	**2.0 (50%)**
Unhandled Rejection	5	2.5 (50%)	3.5 (70%)	**4.5 (90%)**
Resource Exhaustion	4	2.5 (63%)	2.5 (63%)	**3.0 (75%)**
Sequential Awaits	3	2.0 (67%)	2.5 (83%)	**3.0 (100%)**
Event Loop Blocking	3	2.0 (67%)	2.0 (67%)	**2.0 (67%)**
Zalgo	2	1.0 (50%)	1.0 (50%)	**1.0 (50%)**
Context Loss	2	1.5 (75%)	1.5 (75%)	**1.5 (75%)**
Callback Hell	1	1.0 (100%)	0.5 (50%)	**1.0 (100%)**
Stream Leak	1	1.0 (100%)	1.0 (100%)	**0.5 (50%)**
Total	**30**	18.0 (60%)	19.5 (65%)	**22.0 (73%)**

ThreadLearn achieves its strongest results on Unhandled Rejection (90%) and Sequential Awaits (100%)—patterns well-represented in the training data and strongly associated with specific fix templates (`try`/`catch` wrapping, `Promise.all` parallelization). Race Condition improves to 70% for ThreadLearn, up from 50% for both baseline models, demonstrating the value of fine-tuning on race-specific patterns. The hardest categories remain Zalgo (50%) and Double Callback (50%), which require semantic understanding of callback invocation semantics rather than syntactic fix substitution. Stream Leak (50%) is the one category where ThreadLearn underperforms the base model (100%), suggesting stream error-handling is underrepresented in the fine-tuning data.

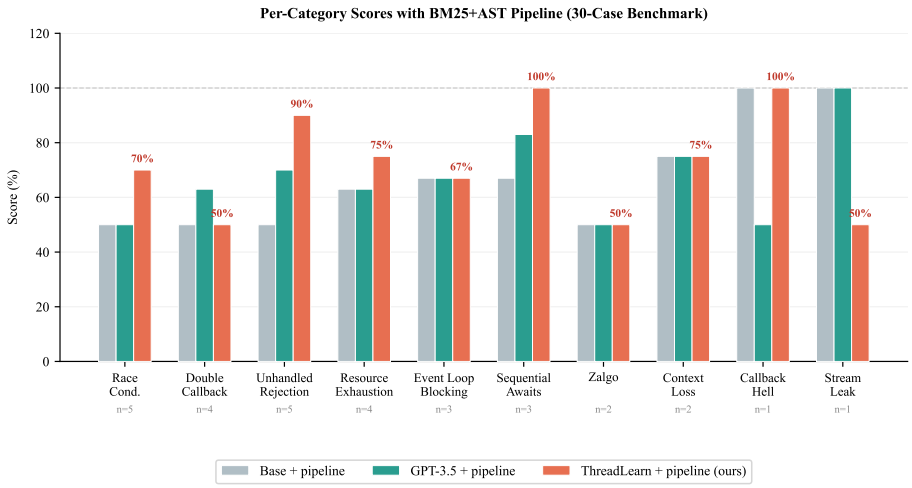

Fig. 4: Per-category scores (%) for all three models with the BM25+AST pipeline on the 30-case benchmark. ThreadLearn (coral) leads in 7 of 10 categories, with perfect scores on Sequential Awaits and Callback Hell. "n=" annotations show the number of test cases per category.

5.6 Inference Latency

Table 14: Observed inference latency per case on Kaggle evaluation runs.

Model	Avg. latency (no pipeline)	Avg. latency (with pipeline)
GPT-3.5-turbo	≈1.8 s	≈1.8 s
Qwen2.5-Coder base	≈17 s	≈18 s
ThreadLearn merged	≈26 s	≈25 s

Local models (Qwen base, ThreadLearn) are ∼13–14× slower than the GPT-3.5-turbo API. This is a cost–privacy trade-off: local models run entirely on-device at no per-call cost with no data leaving the machine. The BM25+AST pipeline adds negligible latency (<0.5 s overhead) to all configurations.

6 Conclusion and Research Directions

Concurrency bugs remain a leading cause of failures in async JavaScript applications. ThreadLearn addresses the gap between static detection and automated remediation by combining a 5-pattern rule-based race detector with a small

language model (Qwen2.5-Coder-1.5B) fine-tuned on 892 hindsight Chain-of-Thought examples and a BM25+AST document retrieval pipeline. On a 30-case real-world benchmark drawn entirely from production npm packages (rw_01–rw_30), ThreadLearn with the full pipeline achieves **22.0/30 (73.3%)**, compared with 60.0% for the base model without fine-tuning and 65.0% for GPT-3.5-turbo with the same pipeline— a gain of +13.3 percentage points over the base model. Crucially, ThreadLearn outperforms GPT-3.5-turbo despite having $\sim$125$\times$ fewer parameters, demonstrating that domain-specialized fine-tuning at 1.5B scale is more effective than general-purpose retrieval augmentation on a larger model. To our knowledge, ThreadLearn is the first end-to-end system that combines a rule-based static race detector with a fine-tuned small language model and RAG for JavaScript concurrency bug remediation—bridging the gap between PCWMs [2] (LLM-only, large model) and rule-based detectors (ESLint, ThreadSanitizer) that cannot generate fixes.

The ablation reveals a key design principle: *retrieval only helps models that already have domain knowledge*. The base model derives no benefit from the pipeline and introduces regression (2 new FAILs), while the fine-tuned model gains +10 pp from the same retrieval. This suggests that pipeline augmentation should be paired with domain-specific fine-tuning rather than applied as a standalone improvement.

Future work will (1) expand the static detector to cover TypeScript AST patterns; (2) add mutex and semaphore patterns for broader concurrency coverage; and (3) investigate whether a larger fine-tuned model (e.g., Qwen2.5-Coder-7B) would push Unhandled Rejection and Race Condition above 75%.

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